

Feature detection: Edges

Canny edge detector

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Outline

- Canny edge detector

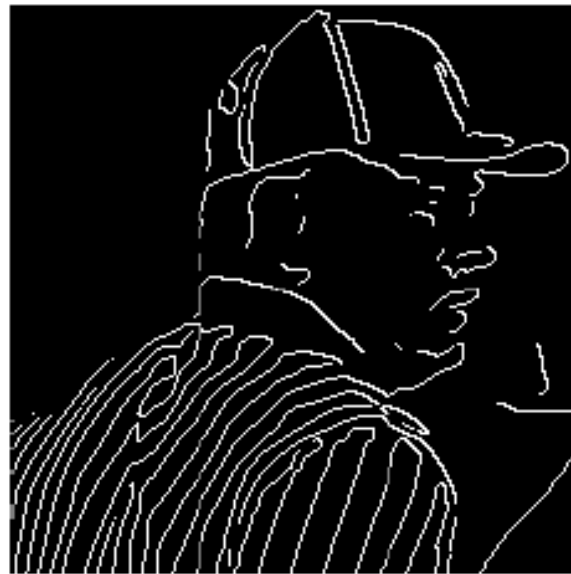


Thresholded 1st driv vs zero-crossing of 2nd driv

$$\|\nabla I\| \geq T \rightarrow$$



$$\Delta I \approx 0 \rightarrow$$



Canny edge detector

- (Probably) The most widely used edge detector
- Proposed by: John Canny, 1986

Canny: Main steps

- Noise filter the image with Gaussian
- Compute magnitude and orientation of gradient
- Suppress non-maximal (i.e. for edge thinning)
- “Connect” broken edges (hysteresis)

1 – Smooth with Gaussian

I



$G_\sigma * I$



2 – Magnitude and orientation

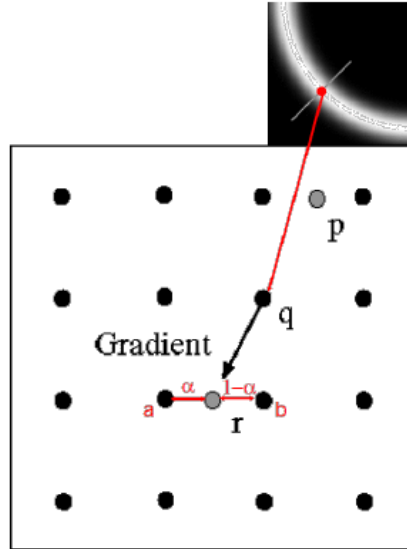
$$\|\nabla I\|$$



$$\arctan\left(\frac{I_y}{I_x}\right)$$



3 – Suppress non-maxima (from the gradient)



Source: Nuno Vasconcelos

- Thin the edges
- Check if q is larger than p and r
- Yes, keeps q unchanged
- Otherwise, $q = 0$
- Basically, check the 2nd derivative along the gradient direction

3 – Suppress non-maxima (result)

$\|\nabla I\|$



NM Suppressed



4 – Threshold hysteresis (connecting broken edges)

- Strong edges: Apply high threshold t_h to the non-maxima suppressed magnitude:
$$\|\nabla I\|_{\text{strong}} = \|\nabla I\|_{\text{NMS}} \geq t_h$$
- Weak edges: Apply low threshold t_l to the non-maxima suppressed magnitude:

$$\|\nabla I\|_{\text{weak}} = t_l \leq \|\nabla I\|_{\text{NMS}} < t_h$$

- Not an edge if $\|\nabla I\|_{\text{NMS}} < t_l$
- “Extend” the strong broken edge along the weak

4 – Threshold hysteresis (result)

$\|\nabla I\|_s + \|\nabla I\|_w$



Canny



Zero-crossing of the Laplacian vs Canny

$\Delta I \approx 0$



Canny



Canny: Algorithm

- Get gradient from smoothed image (or filter with Gaussian derivatives)
- Calculate magnitude and orientation
- Non-maxima suppression:
 - Check if magnitude value is maximum in the gradient direction
 - Set to 0 if not
- Threshold and link edges
 - Define two threshold: high and low
 - If magnitude is greater or equal to high: strong edge; between high and low: weak edge
 - If magnitude is less than low: not edge
 - Extend strong edges by following the weak

Canny edge detection

- Strengths:
 - Localized edges
 - Less broken edges
- Weaknesses:
 - Complex to implement
 - 3 parameters

Summary: Edge detectors

- Thresholded gradient magnitude ($\|\nabla I\| \geq T$):
 - ▶ Easy to calculate (1st order derivatives)
 - ▶ No parameters (besides the size of kernel: Prewitt/Sobel/etc.)
 - ▶ Thick edges
 - ▶ Raising T will, usually, result in broken edges
- Zero-crossing of the Laplacian of Gaussian (LoG):
 - ▶ Similar to the Laplacian, but...
 - ▶ Less sensitive to noise
 - ▶ Has 1 parameter (i.e. σ)
- Zero-crossing of the Laplacian ($\Delta I == 0$):
 - ▶ Localized edges
 - ▶ No parameters (besides the size of kernel)
 - ▶ Very sensitive to noise (2nd order derivatives)
 - ▶ Zero-crossings are not trivial to find
- Canny:
 - ▶ Localized edges
 - ▶ Robust to noise
 - ▶ Not trivial to implement
 - ▶ Requires 3 parameters (i.e. t_l, t_h, σ)